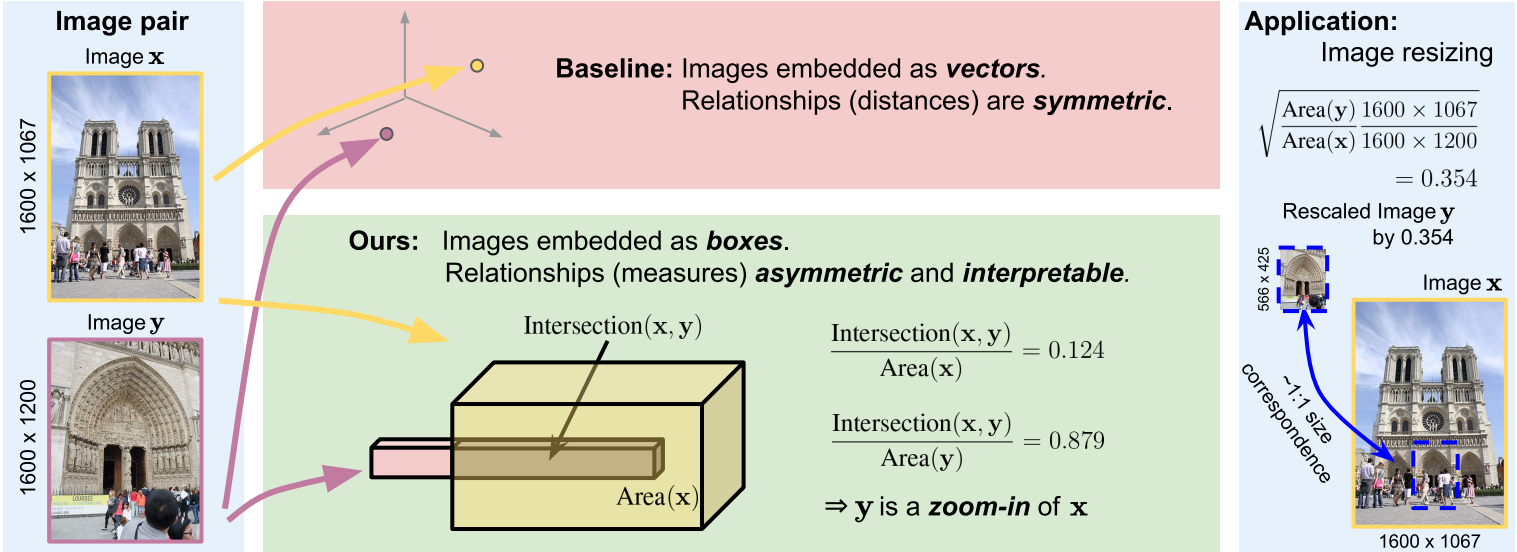
Image pair
Image $\mathbf{x}$
1600 x 1067
Image $\mathbf{y}$
1600 x 1200
Baseline: Images embedded as ***vectors***.
Relationships (distances) are ***symmetric***.
Ours: Images embedded as ***boxes***.
Relationships (measures) ***asymmetric*** and ***interpretable***.
$\text{Intersection}(\mathbf{x}, \mathbf{y})$
$\text{Area}(\mathbf{x})$
$\frac{\text{Intersection}(\mathbf{x}, \mathbf{y})}{\text{Area}(\mathbf{x})} = 0.124$
$\frac{\text{Intersection}(\mathbf{x}, \mathbf{y})}{\text{Area}(\mathbf{y})} = 0.879$
$\Rightarrow \mathbf{y}$ is a ***zoom-in*** of $\mathbf{x}$
Application:
Image resizing
$\sqrt{\frac{\text{Area}(\mathbf{y})}{\text{Area}(\mathbf{x})} \frac{1600 \times 1067}{1600 \times 1200}} = 0.354$
Rescaled Image $\mathbf{y}$
by 0.354
566 x 425
Image $\mathbf{x}$
~1:1 size
correspondence
1600 x 1067